



Numerical modeling of two-phase flow considering multiple bubble sizes in an alkaline water electrolyzer

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ABSTRACT

Precise bubble flow description is crucial for predicting water electrolysis performance. This study employs the Euler-Euler model to simulate bubble flow in an alkaline water electrolysis cell, categorizing the dispersed gas bubble phase into three size classes: 30 μm (small), 90 μm (medium), and 270 μm (large). The volume fraction and velocity fields for each size class were simulated across three current densities, revealing distinct flow patterns: large bubbles exited directly through the outlet, while smaller bubbles were carried along by the circulating liquid. The simulated flow fields were compared with experimentally visualized flow fields, validating the model. Finally, the proposed model demonstrated its advantages over existing models, such as the monodispersed model and the population balance model (PBM), particularly in terms of upward flow velocity near the electrode and bubble descending height in the downward flow.

1. Introduction

Hydrogen, conventionally produced primarily through petroleum refining for various industrial applications, is now drawing attention for its production via water electrolysis due to the growing demand for it as a carrier of renewable energy. However, the proportion of hydrogen produced through water electrolysis remains relatively very low compared to overall production (IEA, 2019). This should be addressed by drastically improving the energy efficiency of water electrolysis and lowering the hydrogen costs. There are three major types of water electrolysis: proton exchange membrane type, alkaline type, and solid oxide type. In order to produce hydrogen with renewable energy, which has a fluctuating and intermittent output, an ideal water electrolysis system must be able to start up in a short time and have a broad load flexibility range (Brauns and Turek, 2020). In terms of that, proton exchange membrane water electrolysis (PEMWE) seems to be the best option. Nevertheless, the major type for large-scale hydrogen production is currently AWE since it offers low investment costs and relies on chloralkali process, a mature traditional technology. While PEMWE uses noble metals like platinum and iridium as its electrocatalysts, AWE uses non-noble metals like nickel and iron, which makes the investment

costs of AWE cheaper than those of PEMWE. Besides, the lifetime of AWE systems is longer and the annual maintenance costs are lower compared to PEMWE systems (Schmidt et al., 2017; Buttler and Spliethoff, 2018). However, AWE needs improvement in its dynamic performance. It is often mentioned that AWE has a long start-up time and a slow dynamic response to the variation of load. Therefore, the dynamic behavior of AWE must be deeply investigated to design and install much larger scales of hydrogen production systems. One of the important factors that affect the dynamic response is the presence of bubbles, which can modify the modality of heat and mass transport (Angulo et al., 2020; Park et al., 2023). Especially for the industrial scale of AWE cells, since they are oftentimes long in the direction of gravity, the volume fraction of bubble in the vicinity of the electrodes can vary considerably depending on location, which makes the electrodes experience heterogeneity and can ultimately lead to electrode degradation. Therefore, it is required to study the influence of the bubble-liquid flow on the dynamic behavior of the electrochemical performance, for which experimental and computational approaches are available. While experimental investigations play a crucial role in understanding the dynamic behavior (Bakker and Vermaas, 2019), conducting experiments with industrial scale AWE cells requires high cost. Hence, a macroscopic computational approach that

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models the behavior of an entire cell or stack is also effective to obtain a comprehensive data about the physical properties of interest. To achieve this, coupling the models of fluid part and electrochemical part is essential. While both independent development of these models and their coupling are crucial, the coupling process requires careful investigation of numerous complex factors. For instance, additional considerations include bubble coverage of the electrode and ion transport through the gaps between bubbles. Hossain et al. (2022) demonstrated that in quasi-steady microscopic simulations that coupled fluid and electrochemical models, the experimental and calculated electric current values often do not align due to the challenges of accurately incorporating these factors. Therefore, in this study, we focused on modeling the fluid part, i.e. bubble-liquid flow. Although some studies have investigated this topic, which is elaborated later in this section, few have effectively simulated the size-dependent bubble flow fields. Generally, there are two main approaches to model macroscopic bubble-liquid flow: separated phase models and dispersed phase models (Fontes, 2024). The separated phase models, which include interface capturing method and interface tracking method, are useful when simulating the detailed surface motion of bubbles. However, for an industrial scale AWE cell, the separated phase models require huge computational cost, thus making the dispersed phase models more desirable. Typical dispersed phase models include Euler-Euler model and Euler-Lagrange model, and when dealing with a large number of bubbles, Euler-Euler model is more suitable.

For the particle size of the dispersed phase, monodisperse size distribution is selected in some cases (Jupudi et al., 2009; Dreoni et al., 2022; Le Bideau et al., 2020). However, since the size variation of electrochemically generated bubbles is so large (Ikeda et al., 2022; Wedin et al., 2003; Boissonneau and Byrne, 2000) that monodisperse models (Le Bideau et al., 2020; Zarghami et al., 2020; Alexiadis et al., 2011) are insufficient to deal with the difference in the behavior of different size bubbles, polydisperse models are more appropriate for the bubble-liquid flow inside a water electrolyzer. As a part of polydisperse models, population balance model (PBM) has been studied by many researchers and various types of PBMs have been developed (Li et al., 2019). The basic concepts behind PBM is to consider the conservation of number density function by solving a population balance equation. Here, number density function represents the number distribution of dispersed particle size (usually volume or diameter). By extracting Sauter diameters from the number density function and use them in the Euler-Euler model CFD simulation, the detailed flow fields for different particle sizes can be estimated. The existing studies of CFD simulations coupled with PBM have provided valid and accurate results in polydisperse systems. However, most of those studies deal only with simple flow configuration such as typical bubble columns as their targets to apply PBM. While bubble column reactors sometimes even have a circulation flow by having components that induce a downward flow, to our best knowledge, no studies have applied PBM to bubble columns with circulation flow (Chen et al., 2004; Bhole et al., 2008; Sanyal et al., 2005; Liu and Tay, 2007). There are some water electrolyzers that have a circulation flow inside them, and since electrochemically generated bubbles have a wide range of size distribution, the velocity fields for each bubble size are often different from one another: small bubbles tend to be carried along by the circulating liquid flow, while larger bubbles tend to rise directly upward without being carried along. Since PBM has only a single momentum equation for the entire dispersed phase, it is not easy to accurately simulate flow systems where there are significant variations in velocity fields depending on the different sizes of the dispersed phase.

Therefore, in this paper, we introduce a polydisperse model that holds multiple size classes, each of which has its own momentum equation so that the different velocity fields are simulated for different size classes. This feature allows for simulating flow fields with significant variations not only in the magnitude but also even in the direction of velocity fields for each particle size class. Using this model, we focused on simulating the bubble-liquid flow field in an alkaline water electrolysis cell.

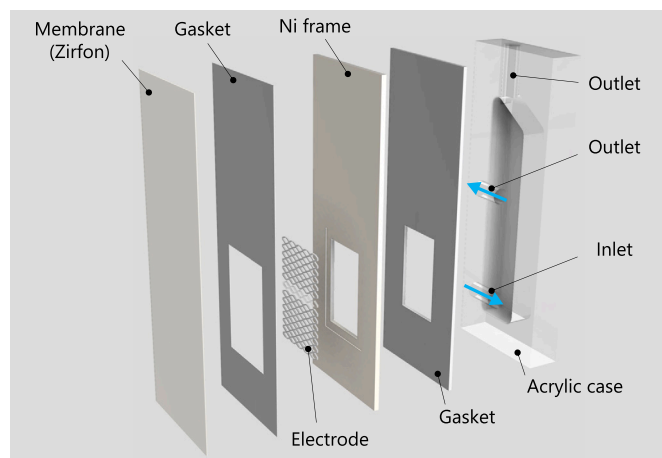


Fig. 1. Configuration of the experimental AWE anodic half-cell, with electrolyte flow indicated by the blue arrows indicate. The smaller electrode is rendered as transparent since it was not in use. Some components are omitted for simplicity.

2. Methodology

2.1. Physical setup

Fig. 1 shows anode side of the alkaline water electrolysis cell used for validation of the presented CFD model. This cell is zero-gap mono-polar type and only its anode side is considered in the simulation, since this study's target is the fluid flow field. The potassium hydroxide solution (KOH) was pumped from the inlet at the bottom of the acrylic case and flows out from the outlet in the middle. Since this cell was primarily designed to investigate local electrochemical behavior, the nickel expanded metal electrode is divided into two parts: the main electrode and a smaller electrode, although the latter was not used in this study. As electric current is applied, oxygen bubbles are generated from the main electrode. The measurements for validation were performed at the following three current densities: $j = 0.3, 0.6, 1.0$ [A/cm²].

2.2. Numerical modeling

2.2.1. Governing equations

The simulation of bubble-liquid flow in an alkaline water electrolyzer in this work was performed using reactingMultiphaseEulerFoam, a multiphase flow solver in OpenFOAMv2006 based on the Eulerian model (OpenFOAM, 2024). This solver can deal with multiple different fluid components to simulate a multiphase flow involving chemical reactions, and the velocity fields can be calculated for each different component. Although this study does not explicitly deal with chemical reactions, this feature can be exploited to calculate the velocity fields for the bubbles of each different size class by considering each bubble size group as a separate component. Thus, in the bubble-liquid flow simulation, when we introduce N classes of bubble size, all we consider are N gas phases and one liquid phase, i.e., a total of $N + 1$ phases. Each of the phases has its own phase fraction field, and they satisfy the following equation:

$$\sum_i^N \alpha_i + \alpha_l = 1, \quad (1)$$

where α is the phase volume fraction, and the subscripts i and l represent the bubble size class and the liquid phase, respectively. The continuity and momentum equations are written as:

$$\begin{aligned} \frac{\partial}{\partial t} (\alpha_i \rho_i) + \nabla \cdot (\alpha_i \rho_i \mathbf{u}_i) &= \dot{m}_i, \\ \frac{\partial}{\partial t} (\alpha_i \rho_i \mathbf{u}_i) + \nabla \cdot (\alpha_i \rho_i \mathbf{u}_i \mathbf{u}_i) &= \mathbf{F}_i, \end{aligned} \quad (2)$$

$$= -\alpha_i \nabla p + \nabla \cdot (\alpha_i \tau_i) + \alpha_i \rho_i g + M_{il} \quad (3)$$

for the bubble size class i , and

$$\frac{\partial}{\partial t} (\alpha_i \rho_i) + \nabla \cdot (\alpha_i \rho_i \mathbf{u}_i) = 0, \quad (4)$$

$$\begin{aligned} \frac{\partial}{\partial t} (\alpha_i \rho_i \mathbf{u}_i) + \nabla \cdot (\alpha_i \rho_i \mathbf{u}_i \mathbf{u}_i) \\ = -\alpha_i \nabla p + \nabla \cdot (\alpha_i \tau_i) + \alpha_i \rho_i g - M_{il} \end{aligned} \quad (5)$$

for the liquid phase l . Here, the momentum equation for the liquid phase is Reynolds averaged and the Reynolds stress is modeled by the standard k - ε model. In these equations, ρ is phase density, $\mathbf{u}$ is phase velocity vector, g is the gravitational acceleration, and $\dot{m}_i$ is a volumetric mass source term that is only active in the computational cells facing the electrode boundary, which is explained in Section 2.2.2. Also, τ is the stress tensor, which is expressed as

$$\tau = \mu \left\{ 2\mathbf{D} - \frac{2}{3} \text{tr}(\mathbf{D}) \mathbf{I} \right\}, \quad (6)$$

where $\mathbf{D}$ is the strain rate tensor described as

$$\mathbf{D} = \frac{1}{2} \left\{ \nabla \mathbf{u} + (\nabla \mathbf{u})^T \right\}. \quad (7)$$

The viscosity μ is defined as $\mu_i = \mu_g^{\text{mol}}$ for the gas phases and $\mu_l = \mu_l^{\text{mol}} + \mu_l^{\text{turb}}$ for the liquid phase, where μ^{mol} is the molecular dynamic viscosity of the phase, and μ^{turb} is the turbulent viscosity (or eddy viscosity). Since we used the standard k - ε model, the turbulent viscosity is estimated as

$$\mu_l^{\text{turb}} = C_\mu \frac{\rho_l k^2}{\varepsilon}, \quad (8)$$

where k is turbulent kinetic energy and ε is turbulent energy dissipation rate.

We considered the momentum transfer between the bubble size class i and the liquid phase as M_{il} in each momentum transport equation, which is the summation of drag (F_D), virtual mass (F_{VM}), lift (F_L), wall lubrication (F_{WL}), turbulent dispersion (F_{TD}) forces:

$$M_{il} = F_D + F_{VM} + F_L + F_{WL} + F_{TD} \quad (9)$$

The drag force F_D results from the viscosity of the continuous phase when the dispersed phase has a slip-velocity against the continuous phase and it acts in the opposite direction of the slip-velocity.

$$F_D = -\frac{3}{4} \frac{C_D}{d_i} \alpha_i \rho_l |\mathbf{u}_i - \mathbf{u}_l| (\mathbf{u}_i - \mathbf{u}_l), \quad (10)$$

where $\mathbf{u}_i - \mathbf{u}_l$ represents the slip-velocity of the bubble class i against the liquid phase l , d_i is the diameter of the bubble class i , and C_D is the drag coefficient which is determined by Schiller and Naumann model (Schiller, 1933):

$$C_D = \begin{cases} \frac{24}{Re_b} (1 + 0.15 Re_b^{0.687}) & (Re_b \leq 1000) \\ 0.44 & (Re_b > 1000) \end{cases} \quad (11)$$

where Re_b is the Reynolds number of the bubble $Re_b = \rho_l |\mathbf{u}_i - \mathbf{u}_l| d_i / \mu_l$. The Schiller and Naumann model is applied to spherical particles. This study focuses on microscale bubbles. Even the largest bubble in this study has an Eötvös number on the order of 10^{-2} , which is sufficiently small to assume that the bubbles are spherical.

The virtual mass (or added mass) force F_{VM} is generated when the dispersed particles are accelerated relative to the continuous phase (Drew et al., 1979). The relative acceleration of a particle results in carrying some volume of the surrounding fluid along with it, which in turn increases the effective inertia of the particle. The virtual mass force is expressed as:

$$F_{VM} = C_{VM} \alpha_i \rho_l \left(\frac{D\mathbf{u}_i}{Dt} - \frac{D\mathbf{u}_l}{Dt} \right), \quad (12)$$

where C_{VM} is the virtual mass coefficient, which is known to be 0.5 for a spherical particle.

The lift force F_L accounts for the transversal motion of the dispersed particles due to the velocity gradients in the continuous phase (Drew and Lahey, 1987), which is expressed as:

$$F_L = C_L \alpha_i \rho_l (\mathbf{u}_i - \mathbf{u}_l) \times (\nabla \times \mathbf{u}_l), \quad (13)$$

where C_L is the lift coefficient, which is set to 0.5.

The wall lubrication force F_{WL} was incorporated to mitigate the overestimation of the gas volume fraction near the wall, where the velocity boundary layer exists around the bubbles. Without this force, the velocity gradients near the wall cause a dominance of the lift force, which makes the bubbles get overly pushed towards the wall. We adopted the wall lubrication model derived by Antal et al. (1991), which is expressed as:

$$F_{WL} = \frac{C_{WL}}{d_i} \alpha_i \rho_l \left| (\mathbf{u}_i - \mathbf{u}_l)_{\parallel} \right|^2 \mathbf{n}_w, \quad (14)$$

where $(\mathbf{u}_i - \mathbf{u}_l)_{\parallel}$ is the bubble relative velocity component tangential to the wall, $\mathbf{n}_w$ is the normal vector of the wall with the inward direction, and C_{WL} is the wall lubrication coefficient:

$$C_{WL} = \max \left(0, C_{w1} + C_{w2} \frac{d_i}{2y_w} \right), \quad (15)$$

where y_w is the distance to the nearest wall, and C_{w1} and C_{w2} are the constant coefficients with adopted values of -0.1 and 0.15 , respectively.

The turbulent dispersion force F_{TD} was employed to include the influence of the turbulent random fluctuations on the redistribution of bubbles in the transversal direction (Lubchenko et al., 2018). We used the model derived by Burns (Burns et al., 2004), which is expressed as:

$$F_{TD} = -\frac{3}{4} \frac{C_D}{d_i} \alpha_i |\mathbf{u}_i - \mathbf{u}_l| \frac{\mu_l^{\text{turb}}}{Sc_{TD}} \left(\frac{1}{\alpha_i} + \frac{1}{1 - \alpha_i} \right) \nabla \alpha_i, \quad (16)$$

where Sc_{TD} is the Schmidt number of turbulent dispersion, which is set to 1.0.

Note that we assumed bubbles do not break up or coalesce in the entire simulation domain. Firstly, the bubbles of interest in this paper are on the microscale and therefore not prone to breaking up easily. Secondly, coalescence does not occur frequently in the bulk, where the spacing between any bubble pairs is often ample (Hreiz et al., 2015; Lee et al., 2005). However, coalescence admittedly occurs in the vicinity of electrode or at the areas where bubbles accumulate, such as free liquid surface or ceilings. As for the vicinity of the electrode, the bubble diameters to be inputted to the simulation are those of bubbles that have already coalesced with other bubbles and detached from the electrode surface (see Section 2.3), so a particular coalescence model does not have to be used. Regarding the areas where bubbles accumulate, since they are densely populated with a high frequency of coalescence events, it is better to have some coalescence model. Nevertheless, modeling bubble coalescence is not within the scope of this paper, so any coalescence is not considered.

2.2.2. Boundary conditions

The simulation geometry is shown in Fig. 2. The uniform velocity boundary condition is applied to very phase at the inlet boundary, which is $\mathbf{u}_i = (-0.032, 0, 0)$ [m/s]. At the outlet boundary, a constant pressure condition is applied. All the other boundaries including the top face of the geometry are set as no-slip wall boundaries. For every phase volume fraction field, zero-gradient condition is applied to all the boundaries.

The volumetric mass source for the bubble class i in Eq. (2) is defined as:

$$\dot{m}_i = c_i \frac{jM}{C_{\text{reac}} F d_{z_{\text{ele}}}}, \quad (17)$$

where j is the current density, M is the molar mass of the product chemical specie, C_{reac} is the number of electrons that is required to produce

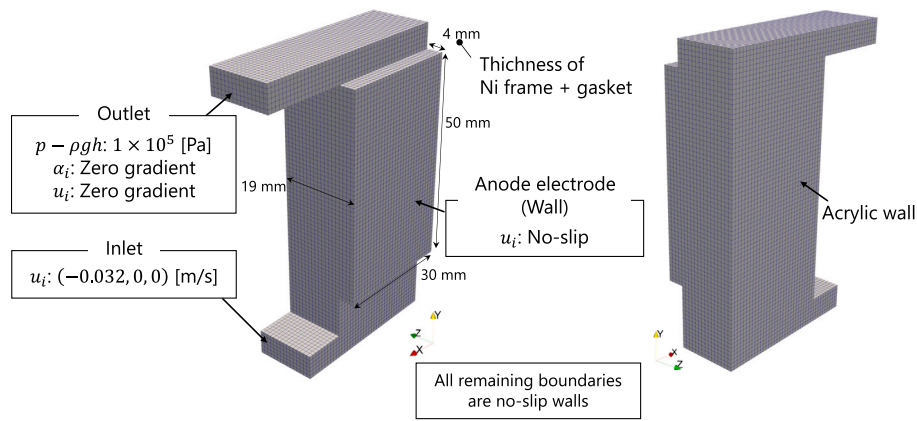


Fig. 2. Simulation geometry of the anodic half-cell, where oxygen is produced, and the boundary conditions for main fields.

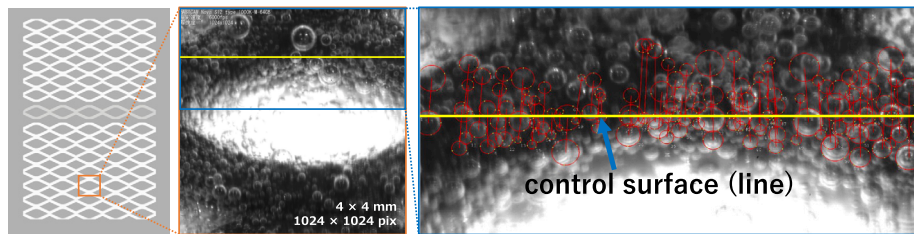


Fig. 3. Measurement of bubble diameter from the experimentally visualized video.

one molecule of the product chemical specie, F is Faraday's constant, and dz_{ele} is the width of the computational cells facing the electrode surface. The fraction composed of these quantities gives the total mass of the chemical species produced by the electrochemical reaction. To allocate this total mass to each bubble size class i , allocation coefficient c_i is multiplied. Accordingly, the sum of c_i must be equal to 1. Note that the mass of water vapor inside bubbles is negligible since the saturation vapor pressure is very low at the temperature of interest, which is 25 °C. In addition, we considered the bubbles' initial velocity at the electrode as zero.

Here, the outlet boundary is extended sufficiently further from the electrode because the pressure boundary at the outlet leads to reverse flow, which causes a violation of Eq. (1). In Fig. 2, however, the outlet is only displayed up to the necessary length.

2.3. Bubble size classes

In this study, three size classes X1, X2, and X3 were considered. The representative bubble diameter d_i for bubble size class $i = \text{X1, X2, or X3}$ and its allocation coefficient c_i were determined based on an experimental visualization. Fig. 3 shows a random frame from the visualized video. The video was recorded at 6000 fps, 1024 × 1024 pixels, and shows the bottom 4 mm square of the electrode, giving a resolution of 3.9 μm/pixel. In the video, a control surface with its normal vector parallel to the direction of gravity is set. The diameters of all bubbles passing through the control surface within a randomly selected 0.02 seconds were recorded. Three sets of this procedure (i.e., 0.06 seconds in total) were performed at each current density.

To obtain the mass of each bubble, equation of state was considered:

$$P_{\text{in}} \frac{4}{3} \pi \left(\frac{d}{2} \right)^3 = \frac{m}{M} RT, \quad (18)$$

where P_{in} is the pressure inside the bubble, d is the bubble diameter measured in the experimental videos, m is the bubble mass, M is the molar mass of oxygen, R is the gas constant, T is the temperature of gas, which is 25 °C. Besides, P_{in} is calculated using the following Young-Laplace equation:

$$P_{\text{in}} = P_{\text{out}} + \frac{2\gamma}{d/2}, \quad (19)$$

where P_{out} is the pressure outside the bubble, γ is the surface tension of alkaline liquid, which is considered to be almost equal to that of water. Eq. (18) and (19) yield the mass m of the bubble with diameter d . Thus, the diameter distribution (Fig. 4a) is converted to the mass distribution (Fig. 4b). Here, to divide the mass distribution into three size classes, the size class boundaries X1-X2 and X2-X3 are set at 60 μm and 180 μm, respectively. The representative diameters of X1, X2, and X3 are set to 30, 90, and 270 μm throughout all the current densities, respectively. With the size class boundaries, the mass allocation coefficient c_i is calculated as the ratio of the total mass of the size class i to the total mass of all the size classes, and they are summarized in Fig. 4c. While it is possible to modify the representative diameters based on conditions such as current density, standardizing the class boundaries and representative diameters offers significant practical advantages in ensuring the general applicability of this model. However, the impact of shifting representative diameters should be investigated, and this is discussed in Section 1.

2.3.1. Grid dependency

Since the grid size affects the simulation results, simulations with varying grid sizes were performed at 0.6 A/cm². Note that the computational geometry consists of a combination of rectangular blocks, and since the grid size is defined for each block, different grid sizes can exist within the entire computational geometry. Therefore, we will refer to each simulation by the total number of grids. Here, in Fig. 5 four simulations with total grid numbers $G = 7410, 18150, 72600$, and 290400 were compared. For each simulation, the y -direction distribution at $z = 0.015$ m (refer to the line A-A' in Fig. 8) on the yz plane was extracted, which was averaged over x direction within the range of $0 \leq x \leq 0.03$ m. From the inset in Fig. 5, the difference between $G = 72600$ and $G = 290400$ is sufficiently small, and considering the balance between accuracy and computational cost, $G = 72600$ was determined to be appropriate.

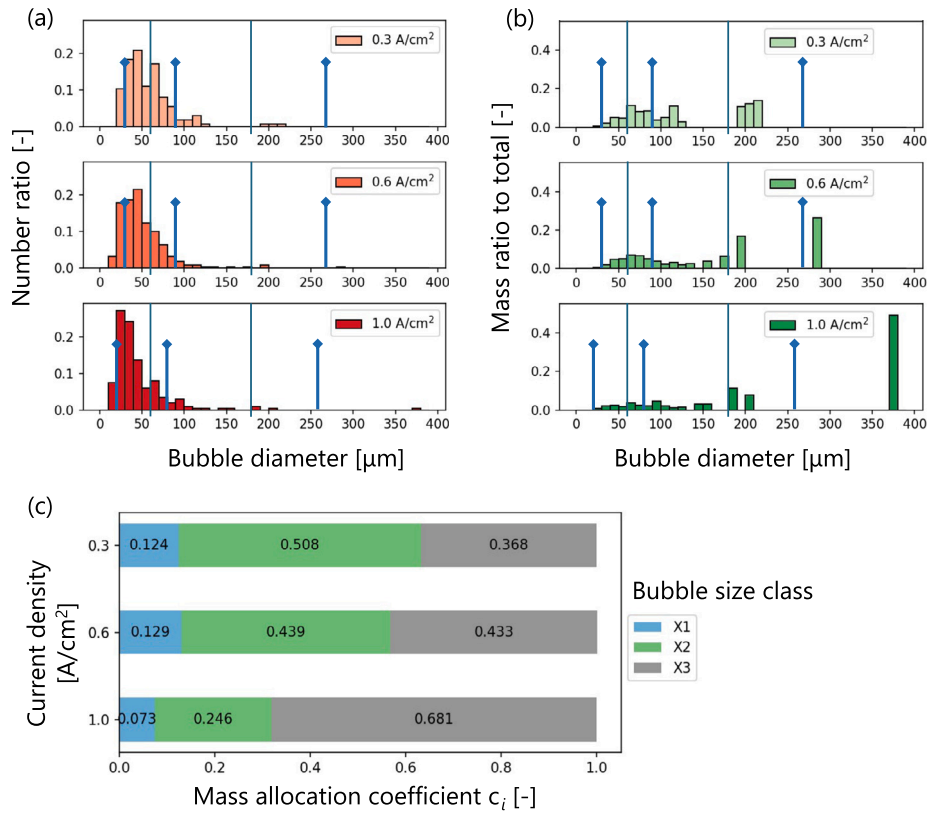


Fig. 4. Histograms of (a) bubble diameter distribution and (b) mass ratio to the total mass. The blue dot lines show the size class boundaries: 60 μm for the boundary between X1 and X2, and 180 μm for the boundary between X2 and X3. (c) The mass allocation coefficients for size classes at each current density.

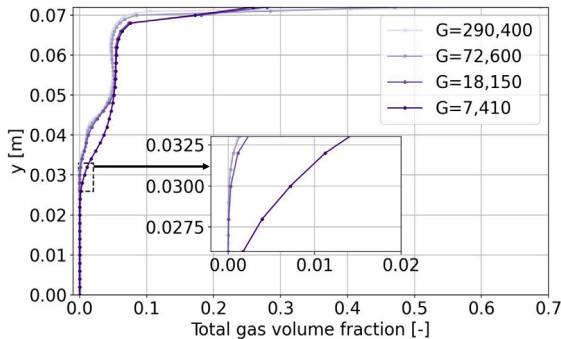


Fig. 5. Grid dependency test with the grid numbers $G = 7410$, 18150, 72600, and 290400. The inset is a magnified plot of the rectangular area outlined by the dashed line.

3. Results and discussion

3.1. The flow field for each of the three bubble size classes

In Fig. 6, the transition of the volume fraction field for each phase at 0.6 A/cm^2 is illustrated (see Supplementary video 1). The electric current was applied on the electrode at $t = 0.0 \text{ s}$ and bubbles started to generate. As all the sizes of bubbles ascend being accelerated by buoyancy force (Fig. 6a, b, and c), the liquid phase is also accelerated due to viscous drag, resulting in a greater velocity of upward flow near the electrode than the average velocity of the inlet-to-outlet flow (Fig. 6e). Consequently, according to the continuity equation, a downward flow occurs on the side of the acrylic wall, leading to the establishment of a circulating flow as a whole (Fig. 6j, and o). The size class X1 was carried by along the liquid downward flow (Fig. 6f), while X3 flowed directly out into the outlet without being carried along by (Fig. 6h), and the flow

field of X2 exhibited characteristics of both X1 and X3 (Fig. 6g). This trend is explained by Stokes number:

$$St = v_t/U_l = \frac{\rho_l d^2 g}{18\mu_l} / U_l, \quad (20)$$

where v_t is the terminal velocity of a bubble, ρ_l is the density of liquid phase, d is the bubble diameter, g is the gravitational acceleration, μ_l is the viscosity of liquid phase, and U_l is the velocity of liquid phase in the downward flow. Here, the values of $\rho_l = 1300 \text{ kg/m}^3$, $g = 9.8 \text{ m/s}^2$, and $\mu_l = 1.63 \times 10^{-3} \text{ Pa s}$ were used. For U_l , the downward liquid velocity at each current density was obtained by averaging the liquid velocity over the area $0.06 \leq y \leq 0.07 \text{ m}$ and $0.015 \leq z \leq 0.018 \text{ m}$ on the yz plane at $x = 0.02 \text{ m}$ (refer to the red rectangular in Fig. 7a). Obtained U_l and Stokes numbers are also shown in Fig. 7a. For every current density, the Stokes number for size class X1 was sufficiently smaller than unity, which explains the trend that the size class X1 followed the liquid downward flow. On the other hand, although the Stokes numbers for size class X3 at the current densities of 0.3 and 0.6 A/cm^2 were sufficiently larger than unity, it was slightly below unity at 1.0 A/cm^2 (Fig. 7b) and it is reflected to its volume fraction field, where a part of the bubbles in X3 is flowing slightly along the liquid downward flow.

As explained thus far, the proposed model is capable of calculating bubble flow fields specific to individual size classes. We believe this capability is highly useful when simulating more complex systems. One such system involves the installation of baffle plates or tubes within the cell. Specifically, as demonstrated by Cai et al. (2024), the installation of baffle plates can enhance the global circulation of the fluid flow, potentially improving electrolytic performance. However, their study only considered a single bubble size, and the effects of incorporating multiple bubble sizes remain unexplored. As mentioned in Section 1, since this study focuses on modeling fluid dynamics at this stage, the quantitative evaluation of these effects is beyond the scope of this paper. Nonetheless, we anticipate that considering multiple size classes will be

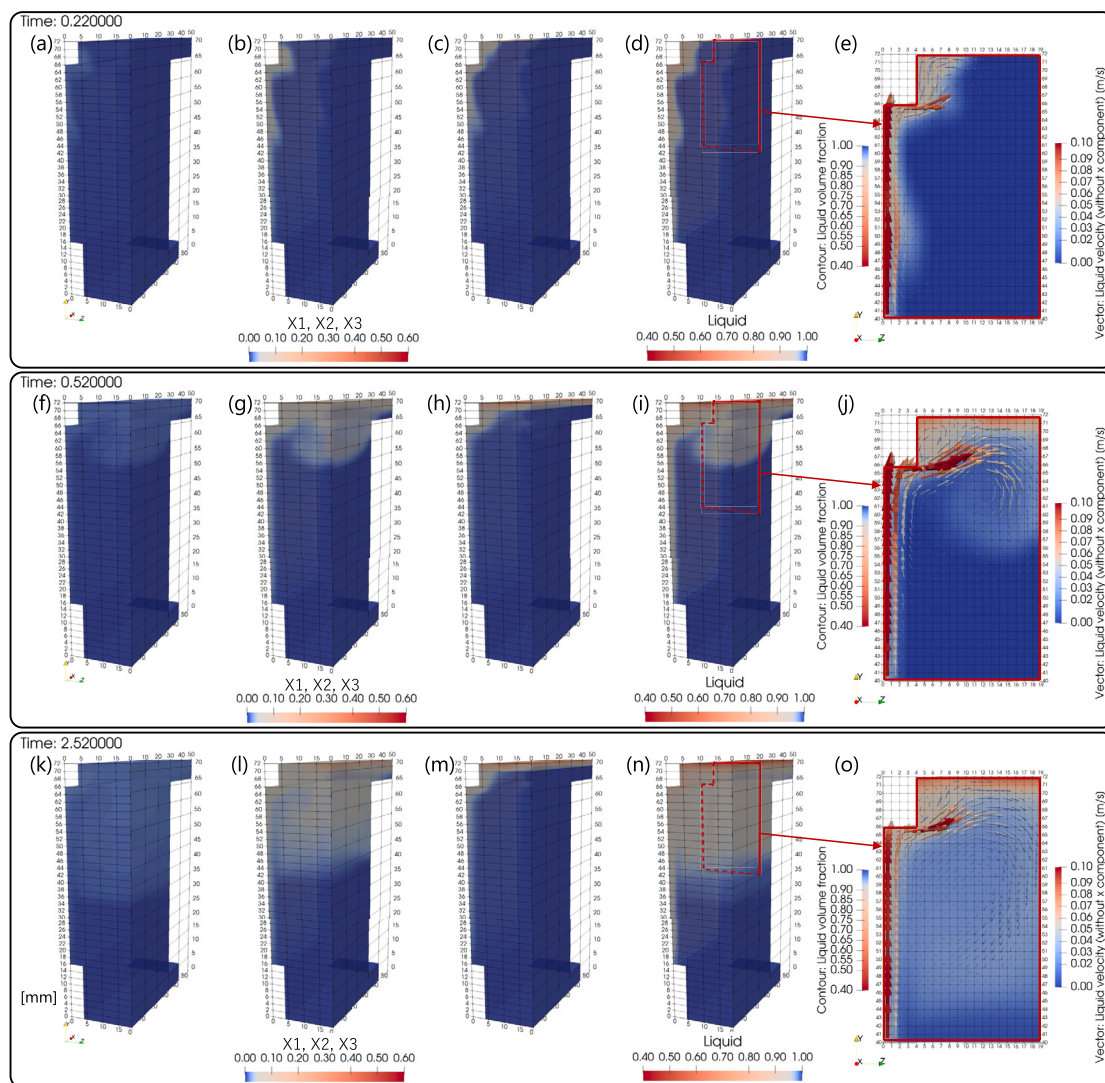


Fig. 6. The phase volume fraction fields for X1 ((a), (f), and (k)), X2 ((b), (g), and (l)), X3 ((c), (h), and (m)), and liquid phase ((d), (i), and (n)) with the extracted liquid velocity fields without x direction component ((e), (j), and (o)) at $t = 0.22, 0.52$, and 2.52 s, respectively, under the current density of 0.6 A/cm^2 .

come even more critical in such complex flow fields. Indeed, as shown in Fig. 9, comparing the pseudo-monodispersed mode, which considers only a single bubble size, with the proposed model, which accounts for multiple bubble sizes, revealed a clear difference in the flow fields. In the corner region at the upper edge of the electrode (corresponding to the rim of the gasket in the experimental cell), the proposed model predicted high velocities toward the direction away from the electrode, whereas the pseudo-monodispersed model did not show such pronounced velocities. Therefore, our proposed model will be effective if this phenomena also occurs in the electrolyzer with baffle plates. (For further discussion about the pseudo-monodispersed model, refer to Section 3.3).

3.2. Model validation by comparing the simulated flow fields with experiments

Among the different current densities, although the similar trend was observed, there was an appreciable difference in the height of the bubble descending flow, which we define as the length measured along y-direction from the top surface of the geometry to the bottom end of the bubble descending flow. This height was compared to that in the experimental visualization as shown in Fig. 8 c. The flow inside the cell was visualized from the side during the steady operation at each of three current densities. In a video, the height of the bubble descending

flow fluctuated, so the 20 seconds of it was time-averaged to a single picture and converted into a gray-scale image. On the other hand, as for the simulated total gas volume fraction field (Fig. 8a), to compare it to the experimental visualization, the total gas volume fraction field at each current density was averaged over x-direction from $x = 0$ to $x = 30 \text{ mm}$. The gas volume fractions (Fig. 8c) from the simulations, and the brightness values (Fig. 8d) from the gray-scale image of the experiment were obtained as the profiles on the line A-A' in Fig. 8b and c. Even though the variation in the brightness values may not always correspond linearly to changes in the total gas volume fraction, since there is still a positive correlation, the height at which the value of interest drops to 20% of its maximum value within the range of $y = 29 \text{ mm}$ to 56 mm was used for quantitative comparison between simulation and experiment. In the experiment, the height of the descending flow increased with the increase in current density. This trend was also observed in the simulations and, and the quantitative values agreed well with the experimental results. However, the increase in the height of the downward flow from 0.6 A/cm^2 to 1.0 A/cm^2 in the simulation is larger than the increase observed in the experiment. In the experiment, bubbles are also discharged from the top surface. The higher current density generates more large bubbles, resulting in the higher upward flow velocity. This leads to more small and medium size bubbles getting trapped into the bubble pool and eventually being discharged from

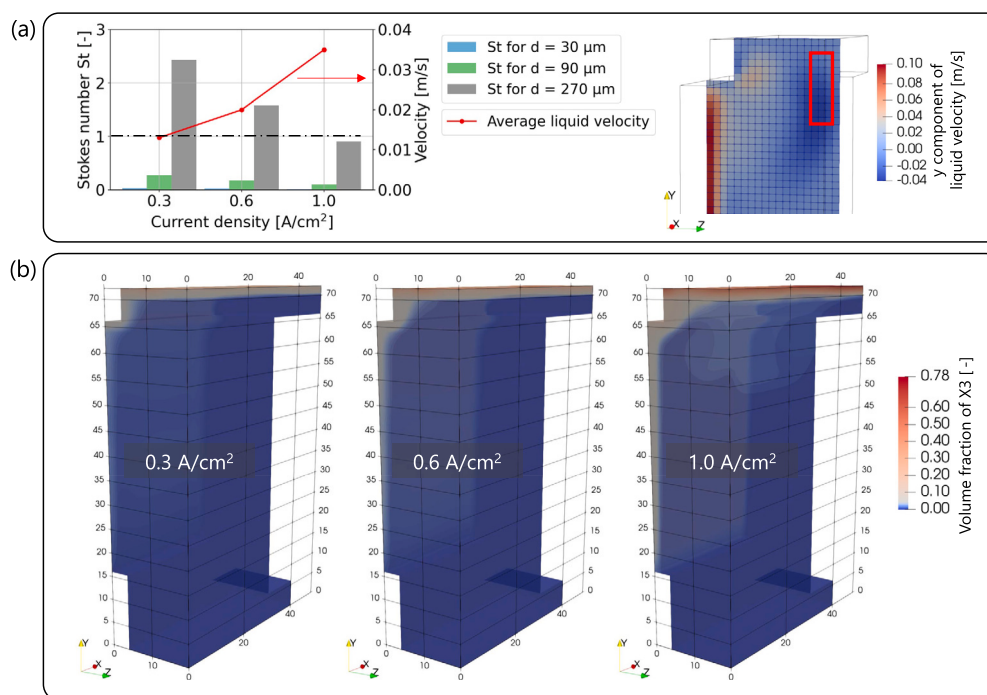


Fig. 7. (a) The downward liquid flow velocity in the simulation, averaged over the red rectangular area, and the Stokes numbers at three current densities. (b) The volume fraction fields of the bubble size class X3 at the three current densities. At 1.0 A/cm^2 , the bubbles of X3 are present in the downward flow as its Stokes number is below unity.

the top surface. Since the presented model does not have such an effect, we consider that even the bubbles that would have been trapped in the bubble pool in the experiment were carried along by the liquid downward flow in the simulation, causing the descending height in the simulation to be larger than that in the experiment. It is worth noting that this deviation is not considered a fundamental issue, although incorporating a bubble coalescence model will be necessary. The experimental cell used in this study was designed to accommodate various experimental conditions, including the use of the top outlet shown in Fig. 1 as the only outlet. However, in practical cells, both the gas phase and the liquid phase are discharged together from a single outlet.

3.3. Advantages of the proposed model over the existing models

The presented model is compared with the main existing approaches for bubble-liquid flow which are mentioned in Section 1: the PBM and monodispersed models. In terms of the velocity field in the upward flow and the descending height of the downward flow, the proposed model offers advantages over both the PBM and monodispersed model. Notably, the flow field near the electrode, where the upward flow occurs, can impact the electrochemical performance.

Since large bubbles constitute the majority of the total bubble mass generated, thereby predominantly influencing the liquid phase velocity field, it is crucial to incorporate large bubbles into a simulation to predict the velocity field in the upward flow accurately. Therefore, with monodispersed models, selecting the diameter of small or medium bubbles as the representative diameter of the bubble size distribution results in an insufficient prediction of the velocity field in the upward flow. To demonstrate this, a pseudo-monodispersed simulation was performed considering only the medium-sized bubbles with a diameter of 90 μm . Fig. 9 shows the sliced velocity field for the liquid and the bubble size class X2 in the simulations with the proposed model and the pseudo-monodispersed model at $t = 2.52$ s. The maximum value of the liquid velocity in the upward flow of the pseudo-monodispersed simulation (Fig. 9b) is smaller than that of the simulation with three sizes (Fig. 9a).

Additionally, as mentioned in Section 3.1, considering the continuity equation, a higher upward liquid flow velocity results in a higher downward liquid flow velocity. Furthermore, due to viscous drag, a higher downward liquid flow velocity leads to a higher downward bubble flow velocity. Therefore, by comparing the downward bubble flow velocity, which can be obtained from the experimental video captured with a high-speed camera, it is possible to indirectly compare the upward liquid flow velocities in the simulations. In the experiment, the bubble flow field near the acrylic wall of the upper part of the electrolyzer was captured (see Fig. 9e, or Supplementary video 2), and the representative velocity of the medium-sized bubbles was measured to be 0.030 m/s by averaging the velocities of 30 medium-sized (90 μm) bubbles. The simulated bubble velocity of the size class X2 within the box area $0.013 \leq x \leq 0.017$ m, $0.058 \leq y \leq 0.062$ m, and $0.015 \leq z \leq 0.018$ m (refer to the red box in Fig. 9), corresponding to the captured area in the experimental video, was averaged. It was calculated to be 0.028 m/s and 0.005 m/s for the three size classes simulation and the pseudo-monodispersed simulation, respectively. While the former value was close to that in the experiment, the latter was significantly smaller than the experimental value. This indicates that the prediction of the upward liquid flow velocity in the pseudo-monodispersed simulation was insufficient. On the other hand, even with the monodispersed model, selecting the diameter of large bubbles as the representative diameter can yield a reasonable velocity field in the upward flow, and needless to say, PBM can provide a better prediction of that.

As for the gas volume fraction in the downward flow, it is required to consider smaller bubbles for an accurate prediction. When selecting the diameter of large bubbles as the representative diameter in the monodispersed model, the gas volume fraction in the downward flow becomes zero, which can be easily inferred from the result that the volume fraction of X3 in the downward flow remained zero (Fig. 6m). On the other hand, the monodispersed model with the diameter of smaller bubbles may simulate the descending height close to that in the experiment as indicated by the volume fraction field of the pseudo-monodispersed simulation (contour in Fig. 9b), despite the velocity in the upward flow (vector in Fig. 9b) did not develop as fast as that of the presented model.

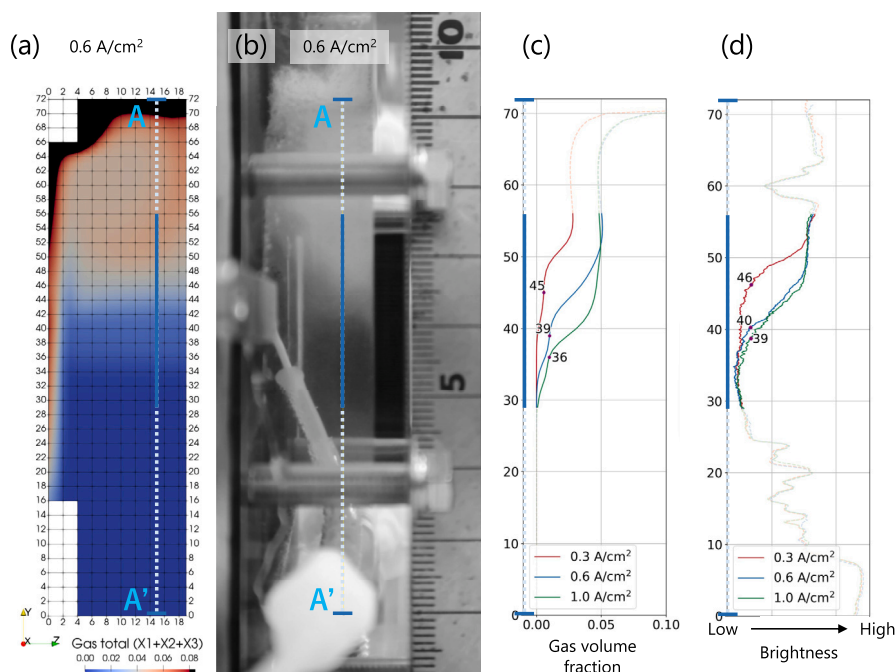


Fig. 8. Comparison of simulation results and the experimental visualization: (a) simulated liquid volume fraction field averaged over the electrode width (30 mm) at 0.6 A/cm^2 , with the volume fraction above 0.08 excluded from color legend, (b) experimental visualization time-averaged over 20 seconds at 0.6 A/cm^2 , (c) the gas volume fraction profiles on line A-A' of simulations at each current density, and (d) the brightness value profiles on line A-A' from the experimental visualizations at each current density.

el's simulation (vector in Fig. 9a). This can be explained by the following two balancing effects. First, an increase in the number of smaller bubbles in the pseudo-monodispersed simulation raises the bubble volume fraction in the downward flow. Second, at the same time, the absence of larger bubbles lowers the upward flow velocity near the electrode, which, in turn, reduces the downward flow velocity, thereby suppressing the volume fraction in the downward flow. PBM models consider smaller bubbles too, so they may be able to simulate the bubbles being carried along by the descending flow. However, they have only one momentum equation for all size classes, which could make it difficult to simulate discrete changes in the bubble size distribution, as in Fig. 6m, where the volume fraction of large bubbles is zero in the downward flow. The proposed model considers multiple bubble sizes, each with its own momentum equation, so that the velocity field in the upward flow near the electrode and the gas descending height of the downward flow are well simulated concurrently.

3.4. Influence of representative diameter choice

As explained in Section 2.3, the representative diameters were fixed for simplicity in this study. Yet, the effects of varying the representative diameters of X1 and X3, which exhibit contrasting characteristics in terms of Stokes number, were investigated. The diameters adopted in this analysis are shown in Fig. 10a. Comparisons of the averaged X2 velocities in the downward flow are presented in Fig. 10b and c, where the cases with X1 varied between 10 and 50 μm and X3 varied among 190, 400, 600 μm are compared alongside the results of the pseudo-monodispersed model and the experimental data. The averaged X2 velocities in the downward flow were calculated over the same range as shown in Fig. 9c. The reason for comparing the velocity of X2 in downward flow is explained in Section 3.3.

As indicated in Fig. 10b, the representative diameter of X1 had minimal impact on the simulated flow field. This is likely because the mass fraction of this class is small, and the Stokes number for X1 remains significantly less than unity in all cases. In other words, the influence of the liquid phase on the bubbles per their volume is strong, resulting in

negligible differences between bubble velocity and liquid velocity. On the other hand, as shown in Fig. 10c, the diameter of X3 has a non-negligible effect. This is attributed to the relatively large mass fraction of this class and its Stokes number being on the order of unity. Larger bubble sizes experience weaker drag force per their volume from the liquid phase, leading to higher bubble rising velocities. However, for bubble sizes larger than 300 μm , the number of bubbles decreases since the total generated mass is constant, thereby reducing the upward force on the liquid phase. Consequently, the overall magnitude of the downward flow weakens, and it was at maximum when the diameter of X3 was 270 μm .

Therefore, admittedly, the effect of the diameter choice for the class X3 was not negligible. However, the pseudo-monodispersed model underpredicted the experimental value by 82%, whereas the model with the representative diameter of 600 μm for X3, which is significantly distant from the obtained bubble size or mass distribution, underpredicted it by only 33%. These results indicate that while the choice of representative diameters does affect the predicted flow field, its influence is limited and does not undermine the conclusion of this study that considering multiple size classes is effective. Having stated this, it still remains important to consider the factors such as the adequacy of obtained sample sizes for the size distributions and changes in them due to current density distributions as future work.

4. Summary

Achieving more precise predictions of bubble flow is crucial for the design of high-performance water electrolyzers. The handling of dispersed bubble size significantly influences the simulated flow fields. While monodisperse models commonly used for predicting bubble flow in water electrolysis cells often yield inadequate predictions, there is a lack of examples utilizing the polydisperse model, such as the Population Balance Model (PBM), for simulating bubble flow including circulating flow inside water electrolysis cells.

In this study, a new model was proposed that classifies the dispersed gas bubble phase into multiple bubble size classes based on bubble diam-

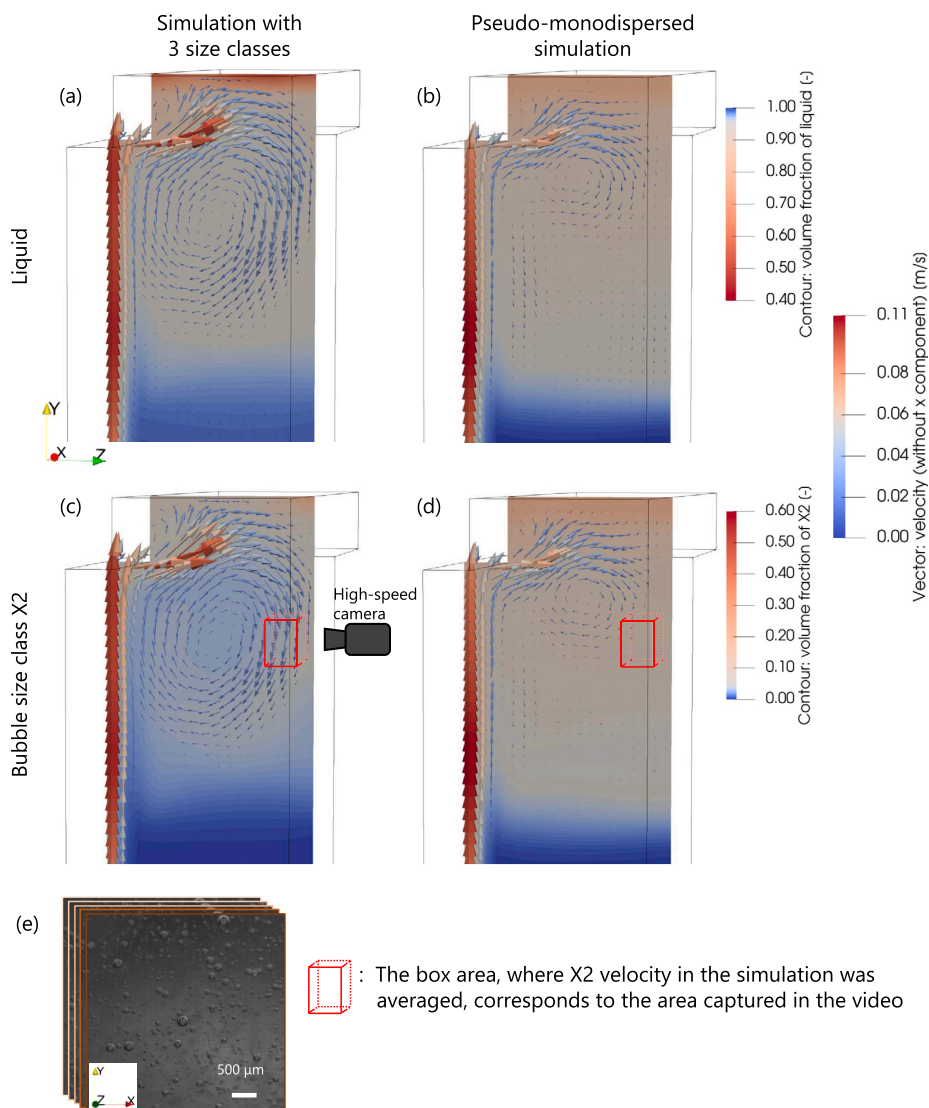


Fig. 9. Volume fraction (color contour) and velocity (vector) fields of liquid ((a) and (b)) and bubble size class X2 ((c) and (d)) on the yz -section at $x = 0.015$ m for the simulations of the proposed model with three size classes ((a) and (c)) and the pseudo-monodispersed model with medium size ((b) and (d)) at $t = 2.52$ s. (e) Experimental video recorded with a high-speed camera to measure the velocity of the medium-sized ($90 \mu\text{m}$) bubble within the area of the red box (see also Supplementary video 2).

eter and calculate individual velocity and volume fraction fields for each size class. This model successfully simulated distinct flow fields of each size class, which is attributed to differences in their Stokes numbers. The simulation results were validated against the experiment for the several current densities. Furthermore, the advantages of the presented model compared to the existing methods, PBM and monodispersed models were illustrated. Future research challenges include formalizing the method for selecting representative diameters for size classes, incorporating a bubble coalescence model, and developing models for microscopic phenomena near the electrode to facilitate coupling with electrochemical models.

CRediT authorship contribution statement

Ryo Kanemoto: Writing – original draft, Validation, Methodology, Investigation, Data curation, Conceptualization. **Takuto Araki:** Writing – review & editing, Supervision, Resources, Funding acquisition, Conceptualization. **Ryuta Misumi:** Resources. **Shigenori Mitsushima:** Supervision, Funding acquisition.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Shigenori Mitsushima reports financial support was provided by New Energy and Industrial Technology Development Organization. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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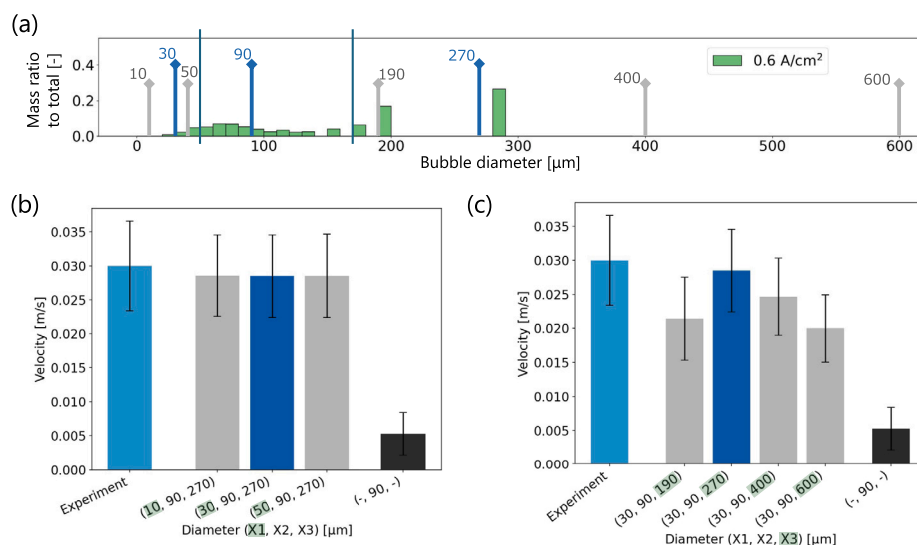


Fig. 10. (a) The adopted representative diameters annotated in the mass ratio distribution (the same distribution shown in Fig. 4b), (b) average velocity of X2 in the downward flow when the diameter of size class X1 was varied, and (c) average velocity of X2 in the downward flow when the diameter of size class X3 was varied. (-, 90, -) indicates the results obtained from the pseudo-monodispersed model.

Appendix A. Supplementary material

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.ces.2024.120986>.

Data availability

Data will be made available on request.

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